FOOD DONATION MARKETPLACE

GROUP PROJECT REPORT

Project: NourishNet – MERN Excess Food Donation Marketplace for NGOs

SUBMITTED BY

1.	Yashashwini Kukkapalli
2.	K. Poojitha
3.	P. Sai Srinidhi
4.	P. Sirivennela

Department / Class	B.Tech – Artificial Intelligence and Machine Learning (AI & ML)
Institution	Malla Reddy Technical Campus
Academic Year	2026–2027

GitHub Repository: https://github.com/yashashwini204/food-donation-marketplace

Prepared from the public repository README. Verify implementation details against the current source code and running application before final submission.

Abstract

The Food Donation Marketplace, named **NourishNet**, is a web application designed to connect organizations with safe surplus food to non-governmental organizations (NGOs), shelters, and charities that can distribute it to people in need. The project describes a MERN-stack implementation with food listings, urgency indicators, NGO reservations, a four-digit one-time password (OTP) for pickup handover, an interactive food-rescue map, role-based workflows, and impact analytics. Its goal is to make surplus-food recovery more organized, traceable, and time-sensitive while encouraging food safety and waste reduction.

Keywords: Food donation, surplus food, NGO, MERN stack, React, Express.js, MongoDB, OTP verification, food rescue, sustainability analytics.

1. Introduction

Restaurants, hotels, caterers, supermarkets, and event venues may have edible surplus food at the end of service. Charities and community shelters need reliable ways to find and collect available food quickly. Informal coordination can make it difficult to track availability, collection times, food-safety information, and successful delivery.

NourishNet proposes a shared digital platform where donors publish food batches and NGOs discover and reserve suitable listings. The workflow includes pickup coordination and confirmation, followed by recording the number of beneficiaries served.

2. Problem Statement

There is a need for a centralized system that helps surplus-food donors and recipient organizations coordinate quickly and transparently. The system should show food availability and urgency, communicate pickup details, reduce unauthorized handovers, and record the social and environmental impact of completed deliveries.

3. Objectives

- Provide a common platform for donors to publish surplus-food batches.
- Help NGOs find and reserve available food based on category, dietary needs, and urgency.
- Coordinate pickup timing and vehicle requirements.
- Use a four-digit OTP workflow to confirm physical handover.
- Record delivery completion and the number of beneficiaries served.
- Display summary metrics related to meals rescued and estimated environmental impact.
- Promote food-safety awareness through declarations and handling guidance.

4. Proposed System

The proposed system has a frontend for user interaction, a backend API for application logic, and a database layer for persistence. The README describes food donor, NGO/charity, volunteer courier, and administrator roles.

Main workflow

- **1. Publish:** A donor adds a food batch with quantity, estimated servings, dietary information, storage details, allergens, and a food-safety declaration.
- **2. Discover:** An NGO views available listings and checks urgency.
- **3. Reserve:** The NGO selects an estimated arrival time, vehicle type, and optional notes, then claims the listing.
- **4. Verify pickup:** A four-digit OTP is checked at handover.
- **5. Complete delivery:** The NGO or volunteer marks delivery complete and records beneficiaries served.
- **6. Review impact:** The application updates displayed meal and sustainability metrics.

5. Major Features and Modules

Module	Description
Food listing and urgent feed	Food batches include category, quantity, estimated servings, dietary classifications, storage temperature, allergen details, and urgency countdowns.
Reservation and OTP handover	NGOs choose an ETA, vehicle type, and notes. A four-digit OTP is used to confirm physical pickup.
Food Rescue Radar	An interactive map displays sample city hubs and color-coded pins corresponding to listing urgency.
Role switcher	Demonstration of donor, NGO/charity, volunteer courier, and administrator views.
Impact analytics	Displays meals rescued, beneficiaries fed, estimated CO■-equivalent emissions prevented, and other environmental equivalents.
Food safety guidance	Includes a donor safety declaration, temperature guidance, and safe transport/reheating procedures.

6. Technology Stack

Layer	Technology	Purpose
Frontend	React 19, Vite	Builds the interactive user interface and supports local development.
Styling	Vanilla CSS	Defines visual theme, responsive cards, and animations.
Backend	Node.js, Express.js	Provides REST APIs and handles application workflows.
Database	MongoDB, Mongoose	Stores application data and supports model-based access.
Authentication	JWT, bcrypt.js	Supports authentication and password hashing as described in the README.

Architecture summary: React/Vite client → Express REST API on Node.js → MongoDB through Mongoose. Authentication and role checks are described as part of the backend design.

7. System Architecture

Presentation Layer React UI — food feed, map, role views, analytics
↓ HTTP / REST API requests
Application Layer Node.js + Express.js — authentication, listings, reservations, OTP verification, delivery completion
↓ Mongoose data access
Data Layer MongoDB — users, food listings, claims, delivery/impact records

8. Key Algorithm / Process Logic

The repository describes a workflow-driven application rather than one standalone algorithm. The core process is:

- Input: donor submits food batch details and safety declaration.
- Validation: backend checks request data and permissions.
- Urgency display: interface shows a countdown based on remaining freshness time.
- Reservation: eligible NGO claims a listing and provides pickup ETA and vehicle details.
- OTP generation: claim receives a four-digit pickup code.
- OTP verification: handover proceeds when the supplied code matches, subject to implemented backend checks.
- Delivery completion: delivery is marked complete and beneficiary count is recorded.
- Impact update: application metrics are updated for display.

Note: Exact validation rules, OTP expiry, retry limits, and impact formulas should be confirmed in source code before presenting them as implemented guarantees.

9. REST API Overview

Endpoint	Method	Purpose
/api/auth/demo-login	POST	Demo sign-in for donor, NGO, volunteer, or admin roles.
/api/auth/register	POST	Register a donor organization or charity.
/api/auth/login	POST	Authenticate with email and password.
/api/listings	GET / POST	Retrieve listings or publish a food batch, subject to permissions.
/api/listings/:id	GET	Retrieve one food listing.
/api/listings/:id/claim	POST	Reserve a food batch and generate pickup OTP.
/api/listings/:id/verify-pickup	POST	Verify pickup OTP for handover.
/api/listings/:id/complete	POST	Complete delivery and record beneficiaries.
/api/claims	GET	Retrieve active claims for the authenticated organization.
/api/analytics/impact	GET	Retrieve impact statistics and leaderboard data.

10. Setup and Execution

Prerequisites

- Node.js version 18 or later (the README notes testing with Node.js v24).
- MongoDB running locally or a configured MongoDB Atlas connection.
- Git and an editor such as Visual Studio Code.

Installation and start-up

- Install backend dependencies: open the server folder and run `npm install`.
- Install frontend dependencies: open the client folder and run `npm install`.
- Configure environment variables locally; do not commit passwords, tokens, or secrets.
- Start the backend using the start script defined in the server package.json (README lists port 5000).
- Start the frontend using `npm run dev` in the client folder (README lists http://localhost:5173).

Security note: Keep real JWT secrets and database credentials private. Verify that no secrets are committed before public release.

11. Testing and Demonstration Plan

The following test cases can be used for a project presentation:

Test	Action	Expected outcome
Browse listings	Open the app and select NGO/Charity role.	Available food batches and urgency details are visible.
Claim food	Choose a listing, select ETA, and confirm reservation.	Claim is recorded and OTP is generated.
Verify pickup	Switch to donor role and submit pickup OTP.	Correct OTP allows handover workflow to proceed.
Complete delivery	Mark delivery complete and record beneficiary count.	Delivery status and beneficiary information are recorded.
Review impact	Open impact analytics.	Impact counters reflect workflow as implemented.

12. Benefits

- Improves visibility of available surplus food and urgency.
- Creates a structured reservation and pickup confirmation process.
- Helps recipient organizations coordinate collection logistics.
- Encourages recording meals distributed and beneficiaries served.
- Raises awareness of food safety and environmental impact.
- Demonstrates interactions among donors, NGOs, volunteers, and administrators.

13. Limitations and Future Enhancements

- Validate authentication, authorization, OTP expiry, rate limiting, and audit logging before real-world use.
- Use real map/geocoding services and validated locations if moving beyond sample hubs.
- Add email, SMS, or push notifications for listings, reservations, and pickup reminders.
- Add image uploads, donor/NGO verification, and moderation tools.
- Document formulas used for estimated CO■ savings and environmental equivalents.
- Add automated tests, deployment monitoring, and backups.
- Review local food-safety and legal guidance before operating the platform.

14. Conclusion

NourishNet presents an approach to coordinating surplus-food donation through a full-stack web platform. Its documented features combine food listing, urgency awareness, NGO reservations, OTP-assisted handover, delivery tracking, and impact analytics. By connecting donors and recipient organizations through a structured workflow, the project aims to make food recovery easier to coordinate and its outcomes easier to monitor. Further testing, security review, and deployment work would be needed before real-world use.

References

- Project repository and README: https://github.com/yashashwini204/food-donation-marketplace
- Technology references: official documentation for React, Vite, Node.js, Express.js, MongoDB, Mongoose, JWT, and bcrypt.js.

Report note: Prepared from the public repository README. Confirm implementation details against the current source code and running application before academic submission.